

Math Symbols List

List of all mathematical symbols and signs - meaning and examples.

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Basic math symbols

Symbol	Symbol Name	Meaning / definition
=	equals sign	equality
≠	not equal sign	inequality
≈	approximately equal	approximation
>	strict inequality	greater than
<	strict inequality	less than
≥	inequality	greater than or equal to
≤	inequality	less than or equal to
()	parentheses	calculate expression inside first
[]	brackets	calculate expression inside first
+	plus sign	addition
−	minus sign	subtraction
±	plus - minus	both plus and minus operations
∓	minus - plus	both minus and plus operations
*	asterisk	multiplication
×	times sign	multiplication
Symbol	Symbol Name	Meaning / definition

MATH SYMBOLS

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$\div$	division sign / obelus	division
$/$	division slash	division
—	horizontal line	division / fraction
mod	modulo	remainder calculation
.	period	decimal point, decimal separator
a^b	power	exponent
a^b	caret	exponent
$\sqrt{a}$	square root	$\sqrt{a} \cdot \sqrt{a} = a$
$\sqrt[3]{a}$	cube root	$\sqrt[3]{a} \cdot \sqrt[3]{a} \cdot \sqrt[3]{a} = a$
$\sqrt[4]{a}$	fourth root	$\sqrt[4]{a} \cdot \sqrt[4]{a} \cdot \sqrt[4]{a} \cdot \sqrt[4]{a} = a$
$\sqrt[n]{a}$	n-th root (radical)	
%	percent	1% = 1/100
‰	per-mille	1‰ = 1/1000 = 0.1%
ppm	per-million	1ppm = 1/1000000
ppb	per-billion	1ppb = 1/1000000000
ppt	per-trillion	1ppt = 10 ⁻¹²



Geometry symbols

Symbol	Symbol Name	Meaning / definition
$\angle$	angle	formed by two rays
$\sphericalangle$	measured angle	
$\sphericalangle$	spherical angle	
$\perp$	right angle	= 90°
°	degree	1 turn = 360°
deg	degree	1 turn = 360deg
'	prime	arcminute, 1° = 60'
"	double prime	arcsecond, 1' = 60"
$\leftrightarrow$ AB	line	infinite line
$\overline{AB}$	line segment	line from point A to point B
$\overrightarrow{AB}$	ray	line that start from point A
Symbol	Symbol Name	Meaning / definition

$\overline{AB}$	arc	arc from point A to point B
$\perp$	perpendicular	perpendicular lines (90° angle)
$\parallel$	parallel	parallel lines
$\cong$	congruent to	equivalence of geometric shapes and size
$\sim$	similarity	same shapes, not same size
Δ	triangle	triangle shape
$ x-y $	distance	distance between points x and y
π	pi constant	$\pi = 3.141592654...$ is the ratio between the circumference and diameter of a circle
rad	radians	radians angle unit
c	radians	radians angle unit
grad	gradians / gons	grads angle unit
g	gradians / gons	grads angle unit

Algebra symbols

Symbol	Symbol Name	Meaning / definition
x	x variable	unknown value to find
$\equiv$	equivalence	identical to
$\triangleq$	equal by definition	equal by definition
$:=$	equal by definition	equal by definition
$\sim$	approximately equal	weak approximation
$\approx$	approximately equal	approximation
$\propto$	proportional to	proportional to
∞	lemniscate	infinity symbol
$\ll$	much less than	much less than
$\gg$	much greater than	much greater than
$()$	parentheses	calculate expression inside first
$[]$	brackets	calculate expression inside first
$\{\}$	braces	set
$\lfloor x \rfloor$	floor brackets	rounds number to lower integer
$\lceil x \rceil$	ceiling brackets	rounds number to upper integer
Symbol	Symbol Name	Meaning / definition

$x!$	exclamation mark	factorial
$ x $	vertical bars	absolute value
$f(x)$	function of x	maps values of x to f(x)
$(f \circ g)$	function composition	$(f \circ g)(x) = f(g(x))$
(a,b)	open interval	$(a,b) = \{x \mid a < x < b\}$
$[a,b]$	closed interval	$[a,b] = \{x \mid a \leq x \leq b\}$
Δ	delta	change / difference
Δ	discriminant	$\Delta = b^2 - 4ac$
Σ	sigma	summation - sum of all values in range of series
$\Sigma\Sigma$	sigma	double summation
$\prod$	capital pi	product - product of all values in range of series
e	e constant / Euler's number	$e = 2.718281828...$
γ	Euler-Mascheroni constant	$\gamma = 0.5772156649...$
ϕ	golden ratio	golden ratio constant
π	pi constant	$\pi = 3.141592654...$ is the ratio between the circumference and diameter of a circle

Linear Algebra Symbols

Symbol	Symbol Name	Meaning / definition
$\cdot$	dot	scalar product
$\times$	cross	vector product
$A \otimes B$	tensor product	tensor product of A and B
$\langle x, y \rangle$	inner product	
$[]$	brackets	matrix of numbers
$()$	parentheses	matrix of numbers
$ A $	determinant	determinant of matrix A
$\det(A)$	determinant	determinant of matrix A
$\ x\ $	double vertical bars	norm
A^T	transpose	matrix transpose
$A^\dagger$	Hermitian matrix	matrix conjugate transpose
A^*	Hermitian matrix	matrix conjugate transpose
Symbol	Symbol Name	Meaning / definition

A	matrix	A^{-1}
$\text{rank}(A)$	matrix rank	rank of matrix A
$\text{dim}(U)$	dimension	dimension of matrix A

Probability and statistics symbols

Symbol	Symbol Name	Meaning / definition
$P(A)$	probability function	probability of event A
$P(A \cap B)$	probability of events intersection	probability that of events A and B
$P(A \cup B)$	probability of events union	probability that of events A or B
$P(A B)$	conditional probability function	probability of event A given event B occurred
$f(x)$	probability density function (pdf)	$P(a \leq x \leq b) = \int f(x) dx$
$F(x)$	cumulative distribution function (cdf)	$F(x) = P(X \leq x)$
μ	population mean	mean of population values
$E(X)$	expectation value	expected value of random variable X
$E(X Y)$	conditional expectation	expected value of random variable X given Y
$\text{var}(X)$	variance	variance of random variable X
σ^2	variance	variance of population values
$\text{std}(X)$	standard deviation	standard deviation of random variable X
σ_X	standard deviation	standard deviation value of random variable X
$\tilde{x}$	median	middle value of random variable x
$\text{cov}(X,Y)$	covariance	covariance of random variables X and Y
$\text{corr}(X,Y)$	correlation	correlation of random variables X and Y
$\rho_{X,Y}$	correlation	correlation of random variables X and Y
$\sum$	summation	summation - sum of all values in range of series
$\sum\sum$	double summation	double summation
Mo	mode	value that occurs most frequently in population
MR	mid-range	$MR = (x_{max} + x_{min})/2$
Md	sample median	half the population is below this value
Q_1	lower / first quartile	25% of population are below this value
Q_2	median / second quartile	50% of population are below this value = median of samples
Symbol	Symbol Name	Meaning / definition

$\bar{x}$	sample mean	average / arithmetic mean
s^2	sample variance	population samples variance estimator
s	sample standard deviation	population samples standard deviation estimator
z_x	standard score	$z_x = (x - \bar{x}) / s_x$
$X \sim$	distribution of X	distribution of random variable X
$N(\mu, \sigma^2)$	normal distribution	gaussian distribution
$U(a, b)$	uniform distribution	equal probability in range a,b
$\exp(\lambda)$	exponential distribution	$f(x) = \lambda e^{-\lambda x}, x \geq 0$
$\text{gamma}(c, \lambda)$	gamma distribution	$f(x) = \lambda^c x^{c-1} e^{-\lambda x} / \Gamma(c), x \geq 0$
$\chi^2(k)$	chi-square distribution	$f(x) = x^{k/2-1} e^{-x/2} / (2^{k/2} \Gamma(k/2))$
$F(k_1, k_2)$	F distribution	
$\text{Bin}(n, p)$	binomial distribution	$f(k) = {}_n C_k p^k (1-p)^{n-k}$
$\text{Poisson}(\lambda)$	Poisson distribution	$f(k) = \lambda^k e^{-\lambda} / k!$
$\text{Geom}(p)$	geometric distribution	$f(k) = p(1-p)^k$
$\text{HG}(N, K, n)$	hyper-geometric distribution	
$\text{Bern}(p)$	Bernoulli distribution	

Combinatorics Symbols

Symbol	Symbol Name	Meaning / definition
$n!$	factorial	$n! = 1 \cdot 2 \cdot 3 \cdot \dots \cdot n$
${}_n P_k$	permutation	${}_n P_k = \frac{n!}{(n-k)!}$
${}_n C_k$ $\binom{n}{k}$	combination	${}_n C_k = \binom{n}{k} = \frac{n!}{k!(n-k)!}$

Set theory symbols

Symbol	Symbol Name	Meaning / definition
$\{ \}$	set	a collection of elements
$A \cap B$	intersection	objects that belong to set A and set B
$A \cup B$	union	objects that belong to set A or set B
$A \subseteq B$	subset	A is a subset of B. set A is
Symbol	Symbol Name	Meaning / definition

$A \subset B$	proper subset / strict subset	A is a subset of B, but A is not equal to B.
$A \not\subset B$	not subset	set A is not a subset of set B
$A \supseteq B$	superset	A is a superset of B. set A includes set B
$A \supset B$	proper superset / strict superset	A is a superset of B, but B is not equal to A.
$A \not\supset B$	not superset	set A is not a superset of set B
2^A	power set	all subsets of A
$\mathcal{P}(A)$	power set	all subsets of A
$A = B$	equality	both sets have the same members
A^c	complement	all the objects that do not belong to set A
$A \setminus B$	relative complement	objects that belong to A and not to B
$A - B$	relative complement	objects that belong to A and not to B
$A \Delta B$	symmetric difference	objects that belong to A or B but not to their intersection
$A \ominus B$	symmetric difference	objects that belong to A or B but not to their intersection
$a \in A$	element of, belongs to	set membership
$x \notin A$	not element of	no set membership
(a,b)	ordered pair	collection of 2 elements
$A \times B$	cartesian product	set of all ordered pairs from A and B
$ A $	cardinality	the number of elements of set A
$\#A$	cardinality	the number of elements of set A
$ $	vertical bar	such that
$\aleph_0$	aleph-null	infinite cardinality of natural numbers set
$\aleph_1$	aleph-one	cardinality of countable ordinal numbers set
$\emptyset$	empty set	$\emptyset = \{ \}$
$\mathbb{U}$	universal set	set of all possible values
$\mathbb{N}_0$	natural numbers / whole numbers set (with zero)	$\mathbb{N}_0 = \{0,1,2,3,4,\dots\}$
$\mathbb{N}_1$	natural numbers / whole numbers set (without zero)	$\mathbb{N}_1 = \{1,2,3,4,5,\dots\}$
$\mathbb{Z}$	integer numbers set	$\mathbb{Z} = \{\dots-3,-2,-1,0,1,2,3,\dots\}$
$\mathbb{Q}$	rational numbers set	$\mathbb{Q} = \{x \mid x=a/b, a,b \in \mathbb{Z}\}$
$\mathbb{R}$	real numbers set	$\mathbb{R} = \{x \mid -\infty < x < \infty\}$
Symbol	Symbol Name	Meaning / definition

$\mathbb{C}$	complex numbers set	$-\infty < a < \infty, \quad -\infty < b < \infty \}$
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Logic symbols

Symbol	Symbol Name	Meaning / definition
$\cdot$	and	and
$\wedge$	caret / circumflex	and
$\&$	ampersand	and
$+$	plus	or
$\vee$	reversed caret	or
$ $	vertical line	or
x'	single quote	not - negation
$\overline{x}$	bar	not - negation
$\neg$	not	not - negation
$!$	exclamation mark	not - negation
$\oplus$	circled plus / oplus	exclusive or - xor
$\sim$	tilde	negation
$\Rightarrow$	implies	
$\Leftrightarrow$	equivalent	if and only if (iff)
$\leftrightarrow$	equivalent	if and only if (iff)
$\forall$	for all	
$\exists$	there exists	
$\nexists$	there does not exists	
$\therefore$	therefore	
$\because$	because / since	

Calculus & analysis symbols

Symbol	Symbol Name	Meaning / definition
$\lim_{x \rightarrow x_0} f(x)$	limit	limit value of a function
ε	epsilon	represents a very small number, near zero
e	e constant / Euler's number	$e = 2.718281828...$
y'	derivative	derivative - Lagrange's notation
y''	second derivative	derivative of derivative
$y^{(n)}$	nth derivative	n times derivation
Symbol	Symbol Name	Meaning / definition

$\frac{d}{dx}$	derivative	derivative - Euler's notation
$\frac{d^2y}{dx^2}$	second derivative	derivative of derivative
$\frac{d^ny}{dx^n}$	nth derivative	n times derivation
$\dot{y}$	time derivative	derivative by time - Newton's notation
$\ddot{y}$	time second derivative	derivative of derivative
$D_x y$	derivative	derivative - Euler's notation
$D_x^2 y$	second derivative	derivative of derivative
$\frac{\partial f(x,y)}{\partial x}$	partial derivative	
$\int$	integral	opposite to derivation
$\iint$	double integral	integration of function of 2 variables
$\iiint$	triple integral	integration of function of 3 variables
$\oint$	closed contour / line integral	
$\oiint$	closed surface integral	
$\oiint$	closed volume integral	
$[a,b]$	closed interval	$[a,b] = \{x \mid a \leq x \leq b\}$
(a,b)	open interval	$(a,b) = \{x \mid a < x < b\}$
i	imaginary unit	$i \equiv \sqrt{-1}$
z^*	complex conjugate	$z = a+bi \rightarrow z^*=a-bi$
$\bar{z}$	complex conjugate	$z = a+bi \rightarrow \bar{z} = a-bi$
$\text{Re}(z)$	real part of a complex number	$z = a+bi \rightarrow \text{Re}(z)=a$
$\text{Im}(z)$	imaginary part of a complex number	$z = a+bi \rightarrow \text{Im}(z)=b$
$ z $	absolute value/magnitude of a complex number	$ z = a+bi = \sqrt{(a^2+b^2)}$
$\arg(z)$	argument of a complex number	The angle of the radius in the complex plane
∇	nabla / del	gradient / divergence operator
$\vec{x}$	vector	
$\hat{x}$	unit vector	
$x * y$	convolution	$y(t) = x(t) * h(t)$
$\mathcal{L}$	Laplace transform	$F(s) = \mathcal{L}\{f(t)\}$
Symbol	Symbol Name	Meaning / definition

δ	delta function	
∞	lemniscate	infinity symbol

Numeral symbols

Name	Western Arabic	Roman	Hebrew
zero	0		
one	1	I	א
two	2	II	ב
three	3	III	ג
four	4	IV	ד
five	5	V	ה
six	6	VI	ו
seven	7	VII	ז
eight	8	VIII	ח
nine	9	IX	ט
ten	10	X	י
eleven	11	XI	יא
twelve	12	XII	יב
thirteen	13	XIII	יג
fourteen	14	XIV	יד
fifteen	15	XV	טו
sixteen	16	XVI	טז
seventeen	17	XVII	יז
eighteen	18	XVIII	יח
nineteen	19	XIX	יט
twenty	20	XX	כ
thirty	30	XXX	ל
forty	40	XL	מ
fifty	50	L	נ
sixty	60	LX	ס
seventy	70	LXX	ע
eighty	80	LXXX	פ
ninety	90	XC	צ
one hundred	100	C	ק

Greek alphabet letters

Upper Case Letter	Lower Case Letter	Greek Letter Name	Letter Name Pronounce
A	α	Alpha	al-fa

$\mathbf{B}$	β	Beta	be-ta
$\mathbf{\Gamma}$	γ	Gamma	ga-ma
$\mathbf{\Delta}$	δ	Delta	del-ta
$\mathbf{E}$	ε	Epsilon	ep-si-lon
$\mathbf{Z}$	ζ	Zeta	ze-ta
$\mathbf{H}$	η	Eta	eh-ta
$\mathbf{\Theta}$	θ	Theta	te-ta
$\mathbf{I}$	ι	Iota	io-ta
$\mathbf{K}$	κ	Kappa	ka-pa
$\mathbf{\Lambda}$	λ	Lambda	lam-da
$\mathbf{M}$	μ	Mu	m-yoo
$\mathbf{N}$	ν	Nu	noo
$\mathbf{\Xi}$	ξ	Xi	x-ee
$\mathbf{O}$	$\omicron$	Omicron	o-mee-c-ron
$\mathbf{\Pi}$	π	Pi	pa-yee
$\mathbf{P}$	ρ	Rho	row
$\mathbf{\Sigma}$	σ	Sigma	sig-ma
$\mathbf{T}$	τ	Tau	ta-oo
$\mathbf{Y}$	υ	Upsilon	oo-psi-lon
$\mathbf{\Phi}$	ϕ	Phi	f-ee
$\mathbf{X}$	χ	Chi	kh-ee
$\mathbf{\Psi}$	ψ	Psi	p-see
$\mathbf{\Omega}$	ω	Omega	o-me-ga

Roman numerals

Number	Roman numeral
0	not defined
1	I
2	II
3	III
4	IV
5	V
6	VI
7	VII
8	VIII
9	IX
10	X
11	XI
12	XII
13	XIII
14	XIV
15	XV
16	XVI
17	XVII
18	XVIII
19	XIX
20	XX
21	XXI
22	XXII
23	XXIII
24	XXIV
25	XXV
26	XXVI
27	XXVII
28	XXVIII
29	XXIX
30	XXX
31	XXXI
32	XXXII
33	XXXIII
34	XXXIV
35	XXXV
36	XXXVI
37	XXXVII
38	XXXVIII
39	XXXIX
40	XL
41	XLI
42	XLII
43	XLIII
44	XLIV
45	XLV
46	XLVI
47	XLVII
48	XLVIII
49	XLIX
50	L
51	LXI
52	LXII
53	LXIII
54	LXIV
55	LXV
56	LXVI
57	LXVII
58	LXVIII
59	LXIX
60	LX
61	LXI
62	LXII
63	LXIII
64	LXIV
65	LXV
66	LXVI
67	LXVII
68	LXVIII
69	LXIX
70	LXX
71	LXXI
72	LXXII
73	LXXIII
74	LXXIV
75	LXXV
76	LXXVI
77	LXXVII
78	LXXVIII
79	LXXIX
80	LXXX
81	LXXXI
82	LXXXII
83	LXXXIII
84	LXXXIV
85	LXXXV
86	LXXXVI
87	LXXXVII
88	LXXXVIII
89	LXXXIX
90	LXXX
91	LXXXI
92	LXXXII
93	LXXXIII
94	LXXXIV
95	LXXXV
96	LXXXVI
97	LXXXVII
98	LXXXVIII
99	LXXXIX
100	C
101	CXI
102	CXII
103	CXIII
104	CXIV
105	CXV
106	CXVI
107	CXVII
108	CXVIII
109	CXIX
110	CXX
111	CXXI
112	CXXII
113	CXXIII
114	CXXIV
115	CXXV
116	CXXVI
117	CXXVII
118	CXXVIII
119	CXXIX
120	CXXX
121	CXXXI
122	CXXXII
123	CXXXIII
124	CXXXIV
125	CXXXV
126	CXXXVI
127	CXXXVII
128	CXXXVIII
129	CXXXIX
130	CXL
131	CXLI
132	CXLII
133	CXLIII
134	CXLIV
135	CXLV
136	CXLVI
137	CXLVII
138	CXLVIII
139	CXLIX
140	CXLI
141	CXLII
142	CXLIII
143	CXLIV
144	CXLV
145	CXLVI
146	CXLVII
147	CXLVIII
148	CXLIX
149	CXLI
150	CL
151	CLXI
152	CLXII
153	CLXIII
154	CLXIV
155	CLXV
156	CLXVI
157	CLXVII
158	CLXVIII
159	CLXIX
160	CLXI
161	CLXII
162	CLXIII
163	CLXIV
164	CLXV
165	CLXVI
166	CLXVII
167	CLXVIII
168	CLXIX
169	CLXI
170	CLXX
171	CLXXI
172	CLXXII
173	CLXXIII
174	CLXXIV
175	CLXXV
176	CLXXVI
177	CLXXVII
178	CLXXVIII
179	CLXXIX
180	CLXXX
181	CLXXXI
182	CLXXXII
183	CLXXXIII
184	CLXXXIV
185	CLXXXV
186	CLXXXVI
187	CLXXXVII
188	CLXXXVIII
189	CLXXXIX
190	CLXXX
191	CLXXXI
192	CLXXXII
193	CLXXXIII
194	CLXXXIV
195	CLXXXV
196	CLXXXVI
197	CLXXXVII
198	CLXXXVIII
199	CLXXXIX
200	C

9	IX
10	X
11	XI
12	XII
13	XIII
14	XIV
15	XV
16	XVI
17	XVII
18	XVIII
19	XIX
20	XX
30	XXX
40	XL
50	L
60	LX
70	LXX
80	LXXX
90	XC
100	C
200	CC
300	CCC
400	CD
500	D
600	DC
700	DCC
800	DCCC
900	CM
1000	M
5000	$\overline{V}$
10000	$\overline{X}$
50000	$\overline{L}$
100000	$\overline{C}$
500000	$\overline{D}$
1000000	$\overline{M}$

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